



Data Mining Course Project, spring 2026

MENTAL HEALTH AND SOCIAL MEDIA

Prepared for:

**Head and TA
of Data mining course**

Prepared by:

Mahmoud Azmy

Marwan Mohamed

Mohmed Adel

Mohand Mesbah

Mohmed Ali

IDs

2406243612

2406249854

2406246737

2406246854

2406241508

01

SUMMARY

Depinning in our data, we try to find out what is the impact of phone usage and social media impact on mental health, where it's more than 92% stressed people coming from it's because of social media.

02

PROBLEM STATEMENT

Try to find what impacts human mental health, and its effect on its daily life, activity, interacting on the day, even its sleep quality.

03

SOLUTION

Turing this report, we are trying to find the most effected habit of acting in daily life for stressed people depending on its phone and social media usage.

PROJECT OVERVIEW

This project presents a complete data mining study conducted on a real-world dataset examining the relationship between social media usage and mental health. The dataset contains 5,000 instances across 15 features, including a mix of numeric and categorical attributes, sourced publicly and relevant to the healthcare and behavioral science domain.

The study is structured across seven interconnected stages. The first stage involves exploratory data analysis and visualization, where the dataset is examined through statistical summaries and seven visualizations covering feature distributions, correlations, and categorical breakdowns. The second stage covers data preprocessing, including outlier treatment, feature engineering, categorical encoding, and feature scaling, with justification provided for every transformation applied.

The third and fourth stages apply unsupervised clustering techniques. K-Medoids clustering is implemented from scratch to identify natural groupings in the data, with the optimal number of clusters determined using the Elbow Method and Silhouette Score. Hierarchical clustering is then applied using three linkage methods – ward, complete, and average – to validate and compare the grouping structure found.

The fifth stage introduces a Fuzzy Logic Inference System designed to assess an individual's mental state. The system takes the cluster assignment from K-Medoids alongside behavioral features as inputs, applies a rule base of nine domain-grounded IF-THEN rules, and produces a mental health evaluation through centroid defuzzification.

The sixth stage applies a Genetic Algorithm for feature selection. The algorithm searches across nineteen candidate features to identify the optimal subset that maximizes clustering quality, measured by silhouette score. The chromosome is encoded as a binary array, and the algorithm uses roulette wheel selection with single-point crossover and bit-flip mutation.

The seventh and final stage integrates all components into a unified implementation pipeline. A single function accepts a new individual's data record alongside the trained models and produces a mental health assessment, demonstrating how the full system operates from end to end.

All implementations are carried out in Python within a Jupiter Notebook environment.

DATASET SELECTION

- We select data coming from open sources (Kaggle). You can check [here](#).
- Which is 5k instances and 15 features.
- Coming from survey

PROJECT TASKS

TASK 1 – EXPLORATORY DATA ANALYSIS

```
In [8]: print("\n3. Statistical Summary : ")
        print(df.describe())
```

3. Statistical Summary :

	age	daily_screen_time_min	social_media_time_min	\
count	5000.000000	5000.000000	5000.000000	
mean	29.947800	373.058200	175.331600	
std	12.279936	106.003916	71.209329	
min	13.000000	140.000000	35.000000	
25%	21.000000	310.000000	118.000000	
50%	27.000000	388.000000	170.000000	
75%	35.250000	461.000000	231.000000	
max	69.000000	520.000000	338.000000	

	negative_interactions_count	positive_interactions_count	sleep_hours	\
count	5000.000000	5000.000000	5000.000000	
mean	0.864200	1.835400	7.134660	
std	0.555176	0.943443	0.533184	
min	0.000000	0.000000	6.400000	
25%	1.000000	1.000000	6.700000	
50%	1.000000	2.000000	7.100000	
75%	1.000000	2.000000	7.450000	
max	2.000000	4.000000	8.300000	

	physical_activity_min	anxiety_level	stress_level	mood_level
count	5000.000000	5000.000000	5000.000000	5000.000000
mean	22.693400	2.510400	7.107600	5.625800
std	10.602862	0.794996	1.062378	0.759928
min	8.000000	1.000000	5.000000	4.000000
25%	14.000000	2.000000	6.000000	5.000000
50%	21.000000	3.000000	7.000000	6.000000
75%	29.000000	3.000000	8.000000	6.000000
max	46.000000	4.000000	9.000000	7.000000

VISUALIZATION

1. SOCIAL MEDIA PLATFORM DISTRIBUTION (PIE CHART)

The pie chart illustrates the distribution of social media platform usage among participants.

Key insight:

The dataset contains multiple platforms with varying usage proportions.

This reflects diverse user engagement across different social media services.

2. MENTAL STATE DISTRIBUTION (BAR CHART)

The bar chart shows a highly imbalanced distribution of mental health states:

Stressed: ~93% (~1,860 participants)

Healthy: ~5.8% (~116 participants)

At Risk: ~1.2% (~24 participants)

Key insight:

The dataset is heavily dominated by individuals classified as "Stressed".

The small proportion of "At Risk" individuals highlight a minority group that may require immediate attention.

This imbalance reflects the dataset's focus on analyzing mental health challenges in relation to digital behavior.

3. GENDER AND PLATFORM USAGE (STACKED BAR CHART)

This visualization shows how different genders interact with various social media platforms.

Key observations:

Certain platforms (e.g., Instagram and TikTok) show higher engagement from specific gender groups.

Messaging-based platforms (e.g., WhatsApp) tend to have a more balanced gender distribution.

Entertainment-heavy platforms may be more associated with specific demographic preferences.

Key insights:

Platform choice is influenced by gender-based behavioral differences.

These differences help explain variations in stress levels observed across platforms.

Demographic factors play an important role in understanding mental health patterns in digital environments.

4. PLATFORM AND MENTAL HEALTH RELATIONSHIP

Across platforms:

The majority of users are classified as "Stressed" in all platforms.

Messaging and communication platforms (e.g., WhatsApp, Facebook) show relatively higher proportions of "Healthy" users.

Visual and entertainment-driven platforms (e.g., TikTok, Instagram, Snapchat) are more associated with stress levels.

Key insight:

Platform type may influence user mental health, with communication-focused platforms potentially being less stressful than short-form content platforms.

5. SCREEN TIME VS AGE GROUP ANALYSIS

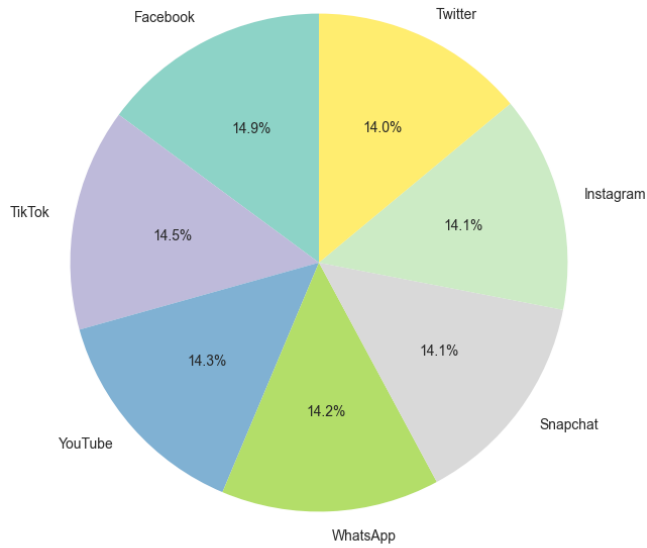
The visualization of screen time across age groups shows:

Younger users tend to have higher screen time usage.

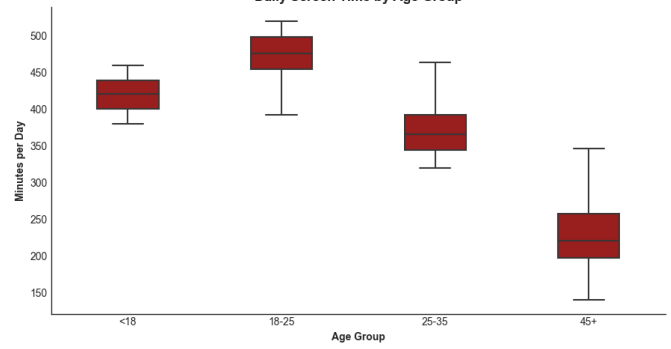
Screen time generally decreases as age increases.

This pattern may explain why younger users are more represented in high-risk clusters.

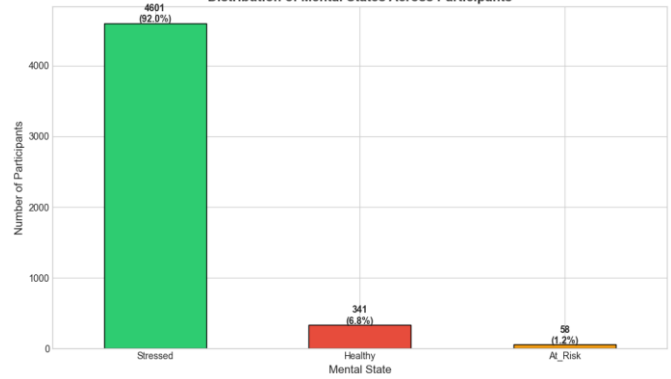
Social Media Platform Usage Distribution



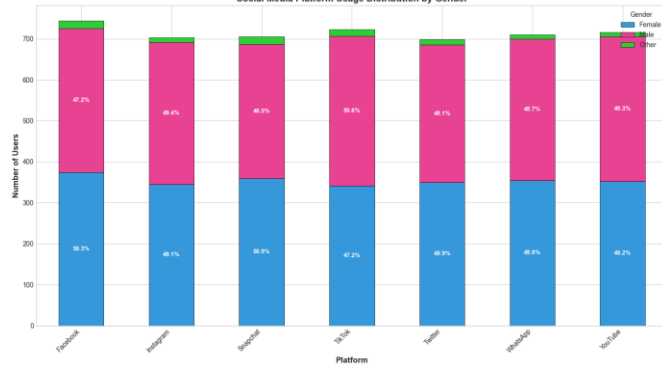
Daily Screen Time by Age Group



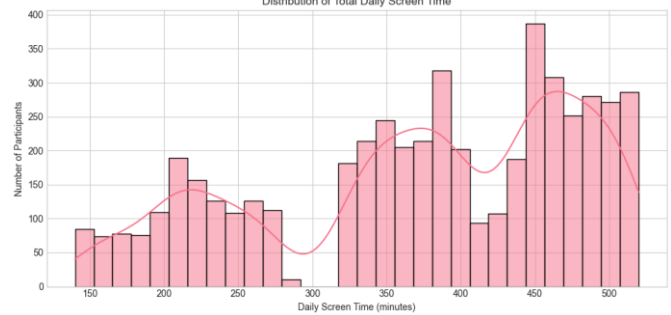
Distribution of Mental States Across Participants



Social Media Platform Usage Distribution by Gender



Distribution of Total Daily Screen Time



DATA PREPROCESSING

This section summarizes all preprocessing transformations applied to the dataset, including the state of the data before and after each step, along with the reason for each transformation.

1. OUTLIER DETECTION (IQR METHOD)

Before:

Raw numerical dataset containing natural variations in age, screen time, sleep, and interaction counts.

After Analysis:

Outliers detected in:

Age: 281 (5.6%)

Negative interactions: 1633 (32.7%)

Positive interactions: 265 (5.3%)

Other variables showed no significant outliers.

After Transformation:

No outlier removal was applied.

Dataset remains unchanged.

Reason:

Age outliers represent valid real-world demographic variation.

Interaction counts are discrete variables, causing false outlier detection due to IQR limitations.

All values are within meaningful and valid ranges; removing them would distort the dataset's distribution.

2. FEATURE ENGINEERING

Before:

Original raw features only (screen time, sleep hours, interactions, etc.).

After:

New features added:

social_media_ratio

net_interaction_score

sleep_deficit

activity_per_screen_hour

Reason:

To capture deeper behavioral relationships between variables.

To improve clustering performance by introducing more informative and meaningful features.

To transform raw measurements into behavioral indicators.

3. CATEGORICAL ENCODING (ONE-HOT ENCODING)

Before:

Categorical variables in text format:

gender

platform

mental_state

After:

Converted into numerical binary columns (One-Hot Encoding):

13 new binary features created

Final dataset size increased to 5000×29

Reason:

K-Medoids clustering is distance-based (Euclidean distance).

Label encoding would create false ordinal relationships between categories.

All categorical variables are nominal with no inherent order.

4. FEATURE SCALING (STANDARDIZATION)

Before:
Numerical features had different scales (e.g., minutes, hours, counts).
After:
Standardized using Z-score normalization:
Mean $\approx$ 0
Standard deviation $\approx$ 1
Reason:
Ensures all features contribute equally in distance-based clustering.
Prevents features with large numeric ranges from dominating the model.

5. FINAL DATASET TRANSFORMATION

Before:
15 original raw features
After:
Expanded to 27-29 features after preprocessing
Final dataset size: 5000 records
Reason:
Feature engineering and encoding increased feature space to improve model expressiveness.
Final dataset is fully numeric and suitable for clustering algorithms.

SUMMARY

All transformations were applied to:
Improve data quality
Ensure compatibility with clustering algorithms
Enhance feature interpretability
Preserve meaningful real-world patterns
The final dataset is clean, standardized, and optimized for machine learning and clustering analysis.

before

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	person_name	age	date	gender	platform	daily_score	social_media	negative_interactions	positive_interactions	sleep_hours	physical_activity	anxiety_level	stress_level	mood_level	mental_status
2	Reyansh Ghosh	35	1/1/2024	Male	Instagram	320	160	1	2	7.4	28	2	7	6	Stressed
3	Neha Patel	24	1/12/2024	Female	Instagram	453	226	1	3	6.7	15	3	8	5	Stressed
4	Ananya Naidu	26	1/6/2024	Male	Snapchat	357	196	1	2	7.2	24	3	7	6	Stressed
5	Neha Das	66	1/17/2024	Female	Snapchat	190	105	0	1	8	41	2	6	6	Stressed
6	Reyansh Banerjee	31	1/28/2024	Male	Snapchat	383	211	1	2	7.1	22	3	7	6	Stressed
7	Myra Kale	25	2/8/2024	Female	Snapchat	516	284	1	3	6.4	8	3	8	5	Stressed
8	Ananya Kulkarni	29	2/19/2024	Other	Snapchat	328	180	1	2	7.4	27	3	7	6	Stressed
9	Meera Das	28	1/11/2024	Female	Facebook	394	138	1	1	7	21	2	7	6	Stressed
10	Vihaan Naidu	31	1/22/2024	Male	Facebook	326	114	0	1	7.4	27	2	6	6	Stressed
11	Meera Patil	26	2/2/2024	Female	Facebook	339	119	0	1	7.3	26	2	6	6	Stressed
12	Suhani Das	39	1/16/2024	Male	WhatsApp	230	58	0	1	7.8	37	1	5	7	Healthy
13	Vivaan Shetty	28	1/27/2024	Male	WhatsApp	363	91	0	1	7.2	24	2	6	6	Stressed
14	Diya Saxena	33	2/7/2024	Female	WhatsApp	376	94	0	1	7.1	22	2	6	6	Stressed
15	Sai Verma	30	1/21/2024	Female	TikTok	387	252	2	2	7.1	21	4	8	5	Stressed
16	Aditi Das	42	1/26/2024	Female	YouTube	223	134	1	2	7.9	38	2	6	6	Stressed
17	Aarav Pawar	24	1/31/2024	Male	Twitter	500	200	1	2	6.5	10	3	8	5	Stressed
18	Avni Rathod	24	2/5/2024	Female	Instagram	456	228	1	3	6.7	14	3	8	5	Stressed

after

	age	daily_screen_time_min	social_media_time_min	negative_interactions_count	positive_interactions_count	sleep_hours	physical_activity_min	anxiety_level	stress_level	mood_level	social_media_ratio	net_interaction_score	sleep_deficit	activity_per_screen_hour	mental_AI_Risk	mental_Healthy	mental_Stressed	gender_I
0	0.411460	-0.500581	-0.215325	0.244632	0.174485	0.497701	0.500538	-0.642080	-0.101292	0.490245	0.215230	0.036125	-0.497701	0.110403	False	False	True	
1	-0.484399	0.754215	0.711613	0.244632	1.234538	-0.815297	-0.725669	0.615914	0.840086	-0.823582	0.206965	1.290473	0.815297	-0.664522	False	False	True	
2	-0.321516	-0.151502	0.290278	0.244632	0.174485	0.122559	0.123243	0.615914	-0.101292	0.490245	0.582331	0.036125	-0.122559	-0.178453	False	False	True	
3	2.936156	-1.727073	-0.987773	-1.556780	-0.885569	1.623128	1.726744	-0.642080	-1.042671	0.490245	0.609380	0.036125	-1.623128	1.938303	False	False	True	
4	0.085693	0.093796	0.500945	0.244632	0.174485	-0.065012	-0.065404	0.615914	-0.101292	0.490245	0.596516	0.036125	0.065012	-0.317881	False	False	True	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
4995	0.981553	-1.123261	-1.563598	-1.556780	-0.885569	1.060415	1.160803	-1.900074	-1.984049	1.808511	-1.642239	0.036125	-1.060415	0.827022	True	False	False	
4996	0.248577	-0.406235	0.543079	0.244632	0.174485	0.497701	0.406214	0.615914	-0.101292	0.490245	1.327210	0.036125	-0.497701	0.029447	False	False	True	
4997	-1.380259	0.262488	1.217215	2.046044	0.174485	-0.252583	-0.254051	1.873908	1.781464	-2.139628	1.339466	-1.218223	0.252583	-0.429209	False	False	True	
4998	-0.728725	0.971210	1.877307	2.046044	1.234538	-1.002868	-1.008640	1.873908	1.781464	-2.139628	1.332264	0.036125	1.002868	-0.777119	False	False	True	
4999	0.981553	-1.170434	-0.187236	0.244632	-0.885569	1.247986	1.160803	-0.642080	-1.042671	0.490245	1.343068	-1.218223	-1.247986	0.866447	False	False	True	

5000 rows x 27 columns

TASK 3: K-MEDOID CLUSTERING

Cluster 0: MODERATE USAGE STRESSED GROUP (32.3%)

Moderate screen time (392 min/day) and social media usage (143 min/day)

Adequate sleep duration (7.0 hours)

Moderate anxiety level (2.22) and stress level (6.91)

Contains only "Stressed" individuals from the original dataset labels

Represents the typical social media user experiencing moderate stress

Cluster 1: HEALTHIER / RESILIENT GROUP (24.9%)

Higher average age (47.3 years)

Lowest screen time (216 min/day) and social media usage (102 min/day)

Lowest anxiety (1.69) and stress levels (5.73)

Highest mood score (6.34) and highest sleep duration (7.92 hours)

Highest physical activity level (38 min/day)

Contains all "At_Risk" (4.7%) and all "Healthy" (27.3%) individuals

Key Insight:

This cluster groups both "Healthy" and "At_Risk" individuals together, suggesting that behavioral patterns derived from social media usage differ from clinical labeling. This may indicate that:

The "At_Risk" label is influenced by external clinical factors not captured in behavioral data, or

Some individuals maintain healthy behavioral habits despite clinical risk classification

Cluster 2: HIGH-RISK / HEAVY USAGE GROUP (42.8%)

Youngest average age (22.4 years)

Highest screen time (450 min/day) and social media usage (242 min/day)

Highest negative interactions (1.21) and also high positive interactions (2.71)

Lowest sleep duration (6.75 hours) and lowest physical activity (15 min/day)

Highest anxiety (3.21) and stress levels (8.06)

Lowest mood score (4.94)

Contains only "Stressed" individuals from the original labels

This cluster represents the group with the highest risk of negative mental health outcomes. Despite relatively high positive interactions, the combination of excessive screen time, high negative interactions, poor sleep, and low physical activity strongly correlates with poor mental well-being.

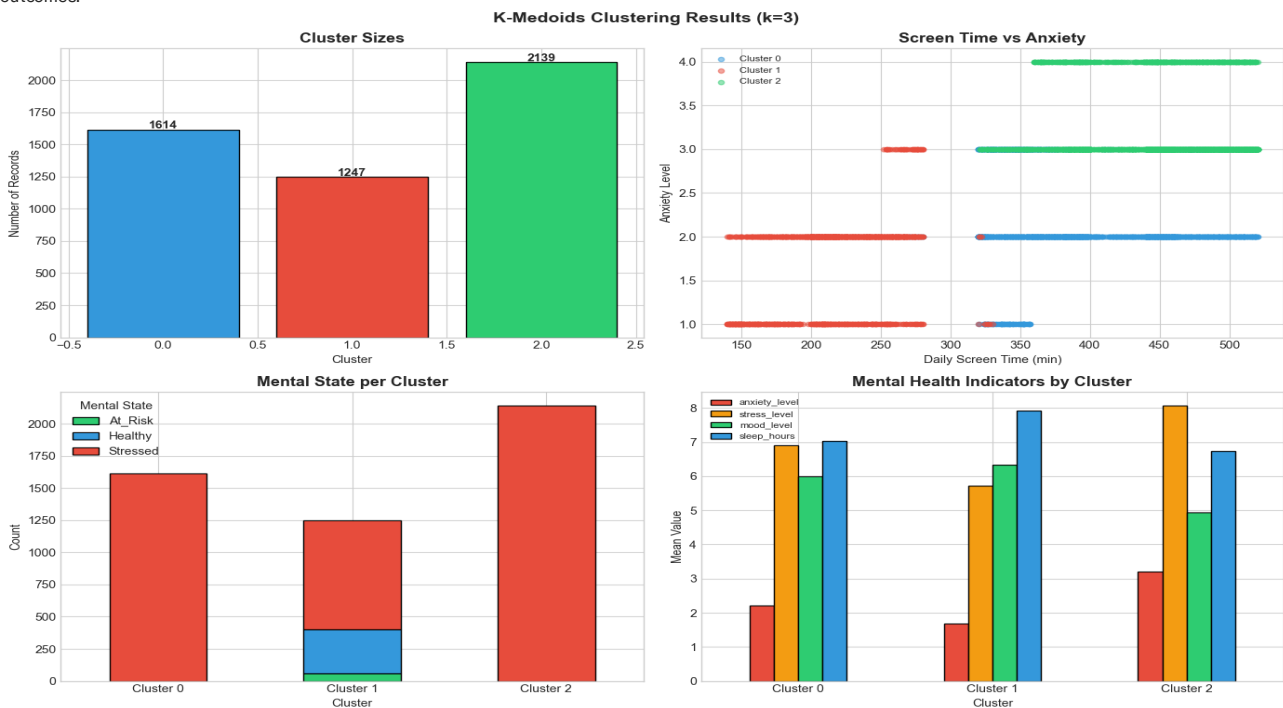
Overall Interpretation

These clusters provide meaningful and actionable groupings for mental health analysis:

Cluster 2 (High-Risk): Requires immediate intervention such as screen time reduction, sleep improvement programs, and stress/anxiety management strategies.

Cluster 0 (Moderate): Suitable for preventive interventions and digital wellness awareness programs.

Cluster 1 (Resilient): Demonstrates that lower social media usage combined with better lifestyle habits (sleep and physical activity) is associated with improved mental health outcomes.



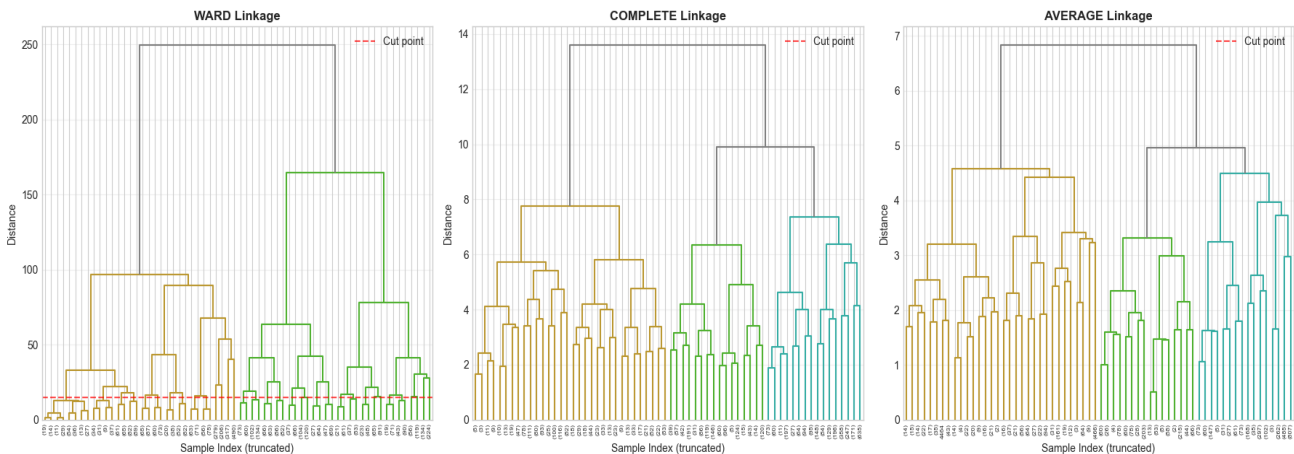
TASK 4: HIERARCHICAL CLUSTERING

KEY FINDINGS

1. K-Medoids and Hierarchical clustering produce highly similar results.
The Adjusted Rand Index (ARI = 0.6824) confirms a strong agreement between both methods, validating the robustness of the three-cluster solution.
2. The clustering structure is primarily driven by the following factors:
 - a. Age (younger individuals tend to cluster together with higher screen time)
 - b. Screen time intensity
 - c. Sleep quality
 - d. Mental health indicators (anxiety, stress, and mood levels)
3. Ward's linkage method generates results similar to K-Medoids because both approaches aim to minimize within-cluster variance, leading to consistent grouping patterns.
4. Hierarchical clustering provides additional interpretability through the dendrogram, which reveals how clusters are formed through successive merging steps.

WHY HIERARCHICAL CLUSTERING ADDS VALUE

- Provides a dendrogram visualization that clearly shows the hierarchy of clusters
- Does not require predefining the number of clusters (k), allowing flexible cluster selection
- Offers a transparent view of how clusters are merged and split during the process
- Serves as a validation method that confirms the findings of K-Medoids clustering



TASK 5: FUZZY LOGIC INFERENCE SYSTEM

Design Overview

A Fuzzy Logic System was developed to estimate Mental Health Risk Score (0-100) using two inputs:

Daily Screen Time (minutes)

Sleep Hours (hours)

The output is a continuous risk score representing a mental health risk level.

K-Medoids clustering was used to validate the fuzzy system by comparing cluster behavior with fuzzy risk outputs.

Why This Model

Screen time and sleep are key behavioral indicators of mental health

Both are measurable in real-world applications

Produces a continuous risk score instead of binary classification

Can be used as an early warning system for mental health monitoring

Fuzzy System Structure

Inputs:

Screen Time: Low / Moderate / High

Sleep Hours: Poor / Adequate / Good

Output:

Risk Score: Low (0–40), Moderate (30–70), High (60–100)

Rule Base (Summary)

Low screen time + good sleep → Low risk

Moderate screen time + adequate sleep → Moderate risk

High screen time + poor sleep → High risk

Sleep quality reduces or increases overall risk level

(9 IF–THEN rules define full system behavior)

Validation Results

Fuzzy system aligns strongly with K-Medoids clusters:

Cluster 1 (Healthy): Lowest risk scores (~26)

Cluster 0 (Moderate): Medium risk scores (~60)

Cluster 2 (High Risk): Highest risk scores (~74)

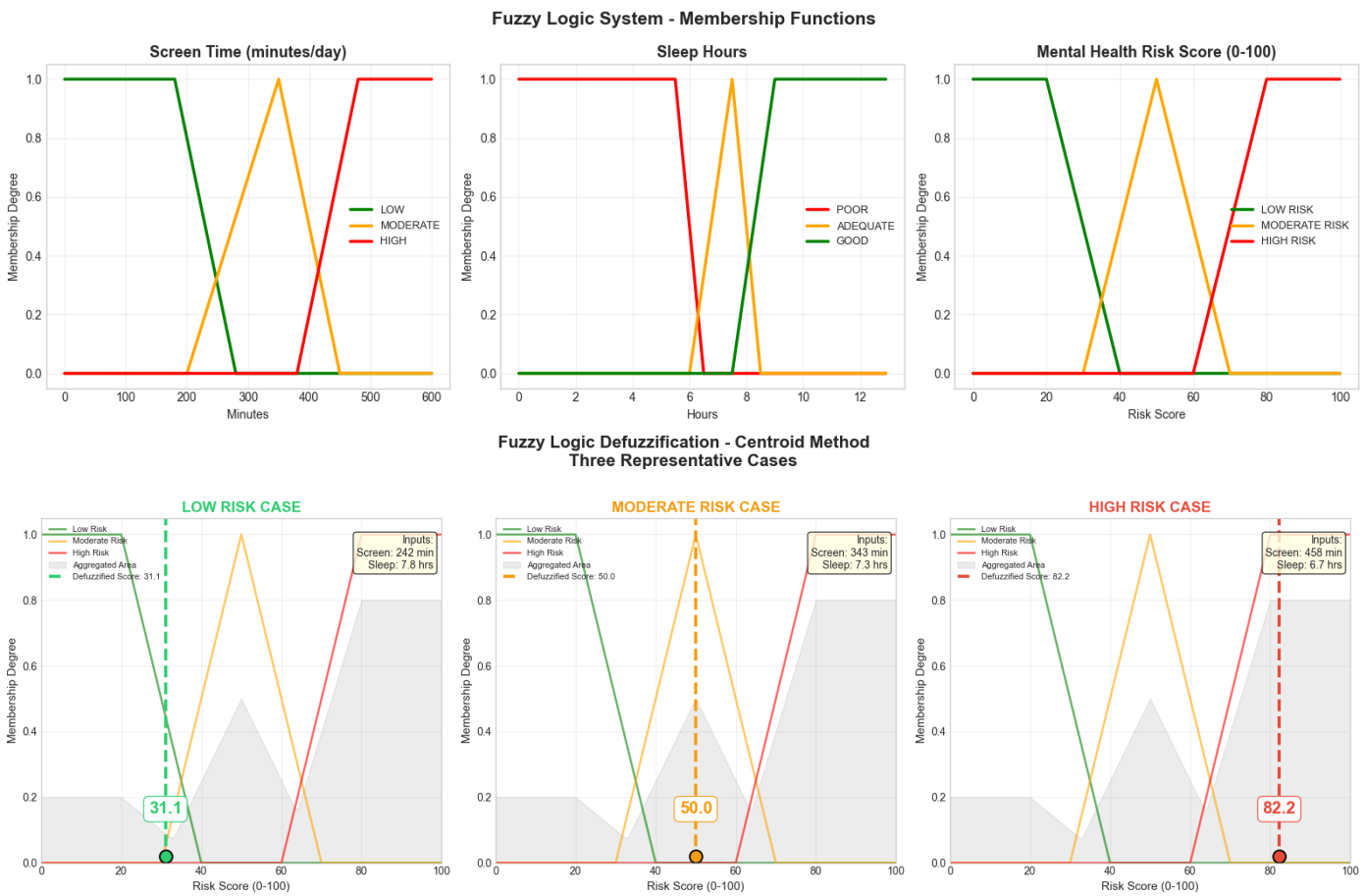
This confirms that the fuzzy system correctly reflects behavioral clustering patterns.

Example Defuzzification Results

Low Risk: 31.1 → Maintain habits

Moderate Risk: 50.0 → Improve sleep or reduce screen time

High Risk: 82.2 → Immediate intervention needed



FEATURE SELECTION USING GENETIC ALGORITHM

PROBLEM DEFINITION

Feature Subset Selection for Mental Health Clustering

OBJECTIVE

Use a Genetic Algorithm (GA) to identify the optimal subset of features that:

Maximizes clustering quality (Silhouette Score)

Minimizes number of selected features

IMPORTANCE OF THIS APPROACH

Identifies the most influential behavioral features for mental health analysis

Reduces data collection cost and complexity

Improves model interpretability

Highlights key factors affecting mental health

Supports future research prioritization

GENETIC ALGORITHM APPROACH

Representation:

Binary chromosome (1 = feature selected, 0 = feature removed)

Fitness Function:

$\text{Fitness} = \text{Silhouette Score} - 0.001 \times \text{number of selected features}$

Genetic Operators:

Selection: Roulette Wheel Selection

Crossover: Single-point crossover (0.8)

Mutation: Bit-flip mutation (0.1)

Elitism: Best solution preserved each generation

Parameters:

Population size: 20

Generations: 30

Minimum features allowed: 3

DATASET INFORMATION

Total features: 24

Numerical: 14

One-hot encoded: 10

Baseline performance (all features):

Silhouette Score: 0.3109

GENETIC ALGORITHM RESULTS

Best solution found:

Selected features: 6 / 24

Feature reduction: 75% (18 features removed)

Best fitness score: 0.5093

SELECTED FEATURES

daily_screen_time_min

social_media_time_min
sleep_hours
stress_level
mood_level
activity_per_screen_hour

REMOVED FEATURES

age
negative_interactions_count
positive_interactions_count
physical_activity_min
anxiety_level
social_media_ratio
net_interaction_score
sleep_deficit
gender (all categories)
platform (all categories)

PERFORMANCE COMPARISON

Metric	Baseline (All Features)	GA Optimized
Number of Features	24	6
Silhouette Score	0.3109	0.5126

CONCLUSION

The Genetic Algorithm significantly improved clustering performance while reducing feature dimensionality by 75%. This confirms that a small subset of behavioral features is sufficient to effectively represent mental health patterns.

EXPLANATION OF THE PIPELINE (PROJECT OVERVIEW)

This project presents a complete data mining study conducted on a real-world dataset examining the relationship between social media usage and mental health. The dataset contains 5,000 instances across 15 features, including a mix of numeric and categorical attributes, sourced publicly and relevant to the healthcare and behavioral science domain.

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IMPLEMENTATION

